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ITA 0517

Computer vision

CO-2

AT3-CLASS TEST

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- 25) filtering Low-frequency Background and High frequency noise.
- 26) Edge Enhancement without Significantly Amplifying noise?
- 27) poor separation Between object and Background.
- 28) Regions with similar intensity but different Textures
- 29) noisy and fragmented detected edges?

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Answers

25) Defination :-

The low-frequency is used for the Background changes and High frequency is used for the to reduce noise of an image. A suitable approach is used for low-pass filter. Such as Gaussian filter.

Working :-

- * Low frequency represents slowly to change background of information.
- * High frequency represents slowly to change the reducing of noise of an image.
- * And the Gaussian filter is used for the Smoothing of an image. and reducing noise of an image.
- * The filtering of an moderate image σ is used for to reduce excessive blur in image.

Justification :-

Gaussian filter is mainly used for mainly smoothing of an image and to reduce the noise of an image. And the low frequency for background information and high frequency mainly for noise.

Conclusion :-

finally using of Low frequency and high frequency of filtering is used for background information and noise of an image. Gaussian for smoothing of an image.

26)

Definition :-

A suitable of approach of an Laplacian of Gaussian, (LoG) filter are using in this edge enhancement.

Working :-

- * Gaussian filtering is used for the image smoothing of an image.
- * In this edge enhancement we are using LoG $\rightarrow$ Laplacian
- * To detect the edges of an image.
- * After detecting edges of an images for that we are going to reduce noise and smoothing the image.
- * And then at last we sharp intensity the images.
- * finally detected edges are going fetch in the image.

Justification :-

In this edge enhancement we are using the (LoG) $\rightarrow$ Laplacian of Gaussian filtering for better smoothing and reducing noise of an image.

Conclusion :-

So finally edges are detected. and then enhance the images as smoothing and sharp intensity and replace to remain visible of an image.

27) Defination :-

A suitable approach is used, morphological filtering for the poor separation b/w object and background.

Working :-

- * morphological filtering used for mainly represents slowly to the changes of separation and background.
- * first we are going to take poor separation object of an image
- * After that the object of image for that sharp intensity and reduce noise and amplify it.
- * And then background of an image for that we can use low frequency for background and for that background of an image we can have to Gaussian filtering for better smoothing and enhancement of an image.
- * finally as for object of an image sharp's intensity and reduce noise and for background changes slowly.

Justification :-

- * in this poor separation b/w object and background we used morphological and low frequency for background information and Gaussian for smoothing.

Conclusion :-

So finally we had reduced the poor separation b/w object and background of an image using morphological and low-frequency and then gaussian filtering.

29) Definition :-

A suitable approach for the noisy and fragmented Detected edges we used High-frequency, Gaussian, LOG (Laplacian of Gaussian).

Working :-

- * In this noisy of an image we are using high frequency to represent slow changes in image.
- * And for the fragmented Detected edges we are using LOG $\rightarrow$ Laplacian of Gaussian filtering.
- * First we take the edges of an image for the better enhancement.
- * After that for the enhanced edges for that using high frequency we are going to reduce noisy.
- * Finally we reduce the noisy and fragmented Detected edges of an image.

Justification :-

for the noisy $\rightarrow$ high-frequency represents slow change the noisy. for the fragmented Detected edges $\rightarrow$ Laplacian of Gaussian used to Detect edges (LOG)

Conclusion.

Finally we reduce the noisy and fragmented detected edges of an image.